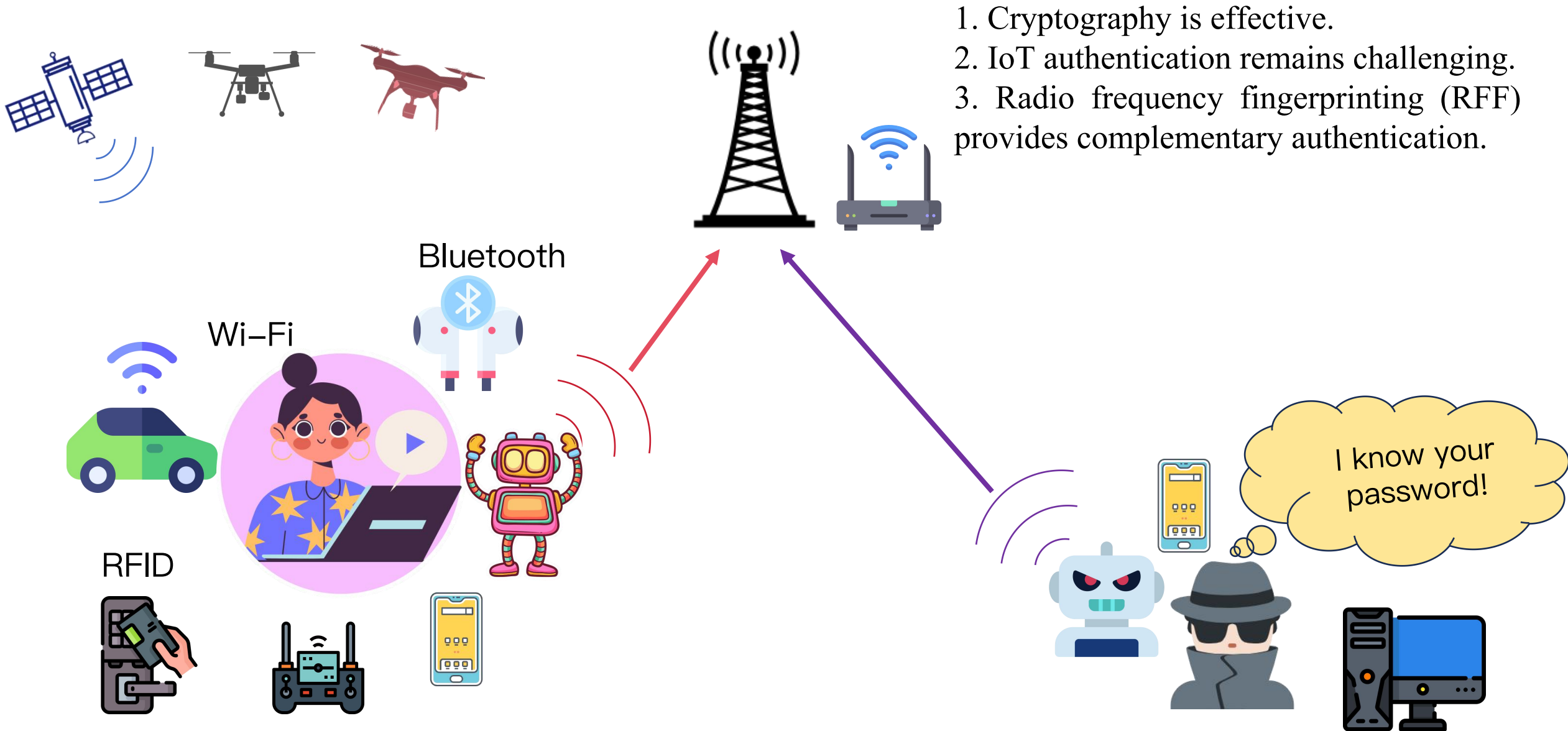
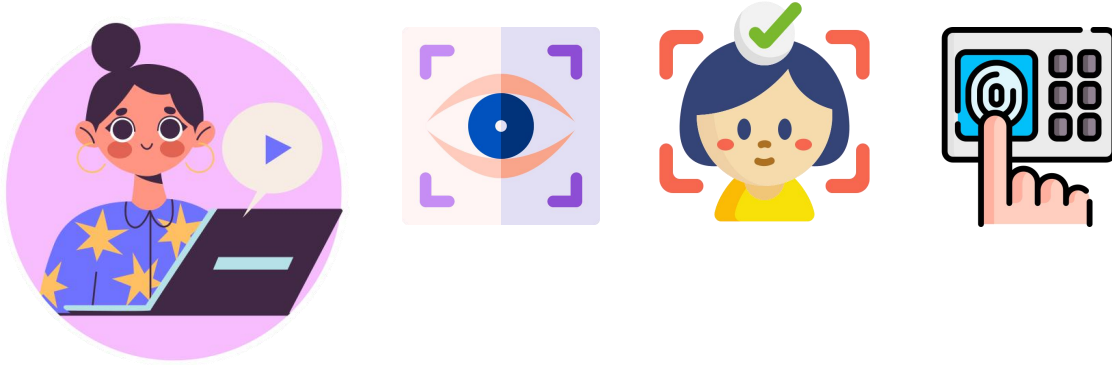


Authentication in our daily life

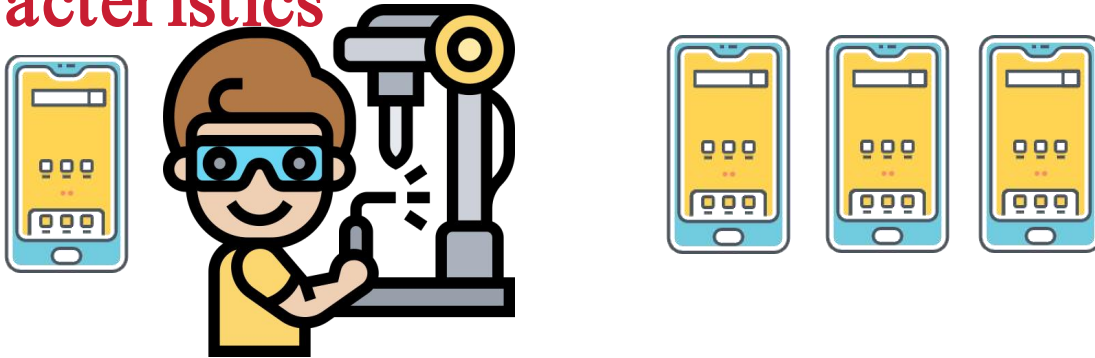


RF fingerprinting (RFF)

Humans have unique biological characteristics

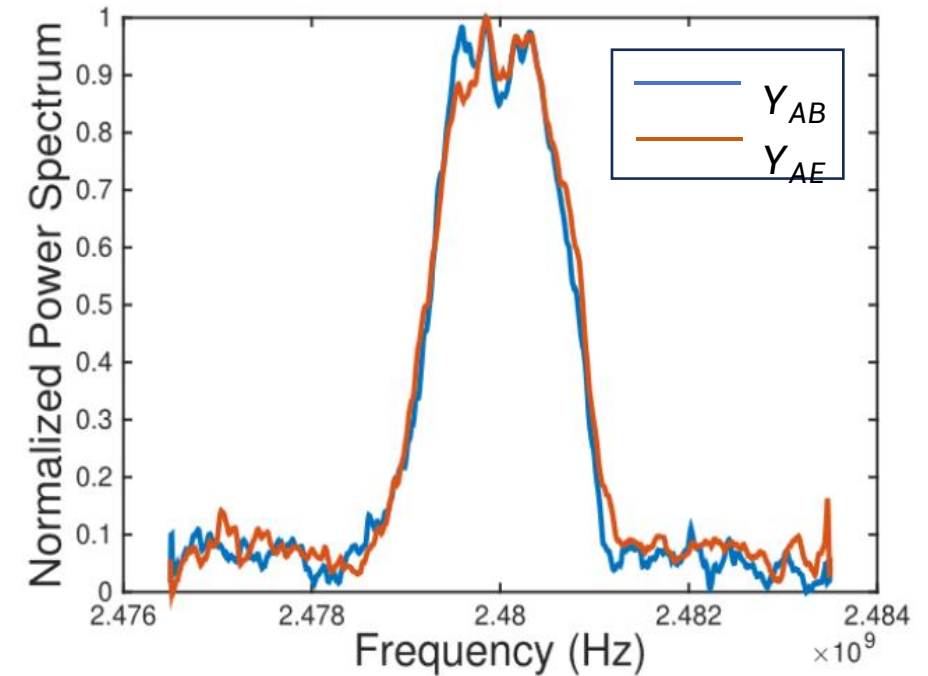
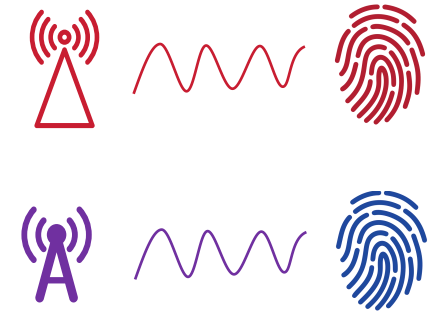


Wireless devices also have unique RF characteristics



Thermal noise,
random vibration,
Hardware non-idealities.

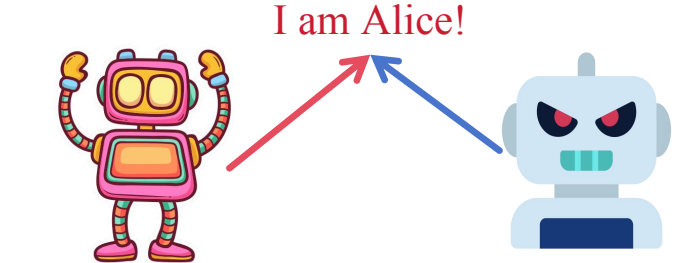
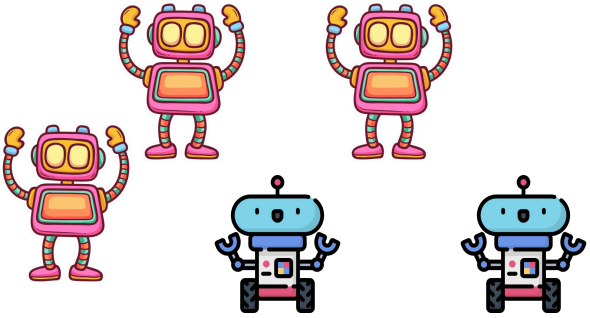
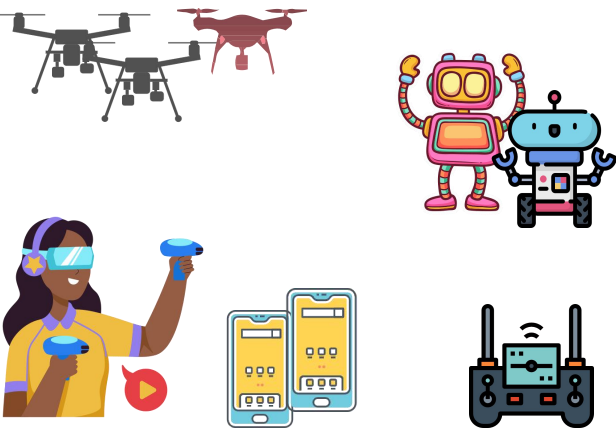
All components are
imperfect.



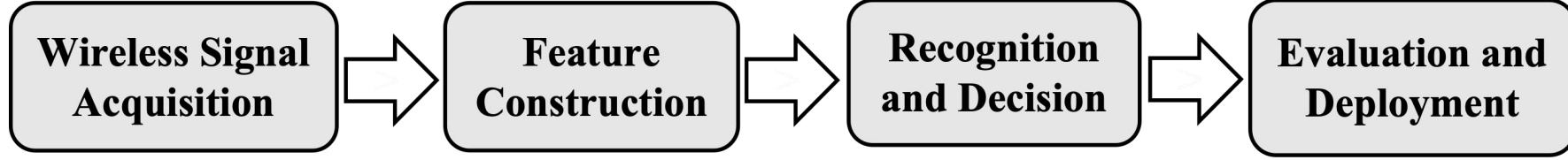
Their signals also have unique
differences.

The applications of RFF

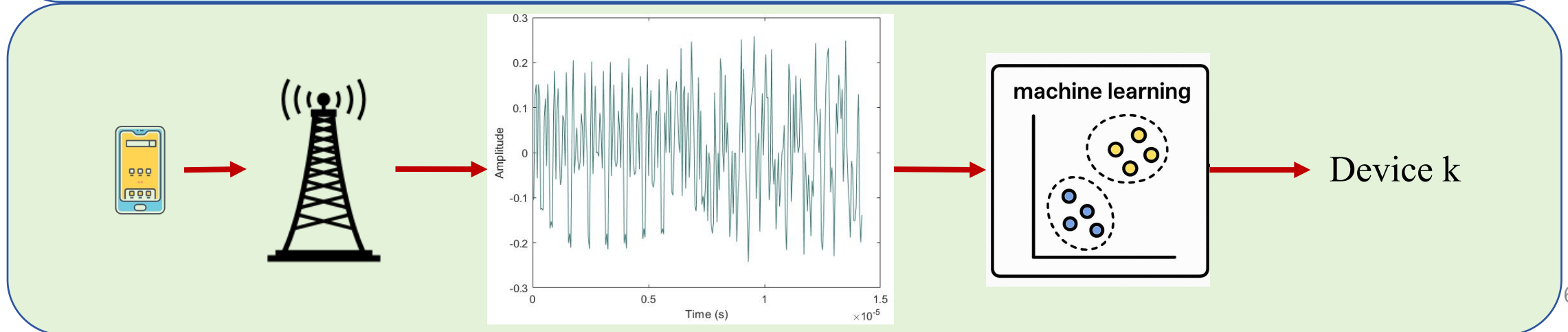
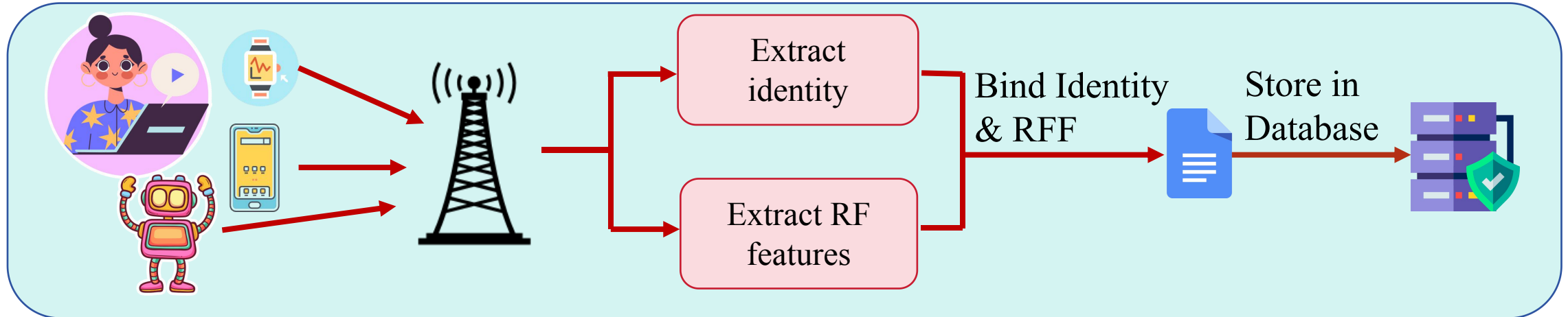
Examples:

Verification	Identification	Classification
 Legitimate transmitter Alice Impersonator Eve		
Is this really Alice?	Which device is transmitting?	What type of device is transmitting?
Accept / Reject	Device ID	Device Type / Model

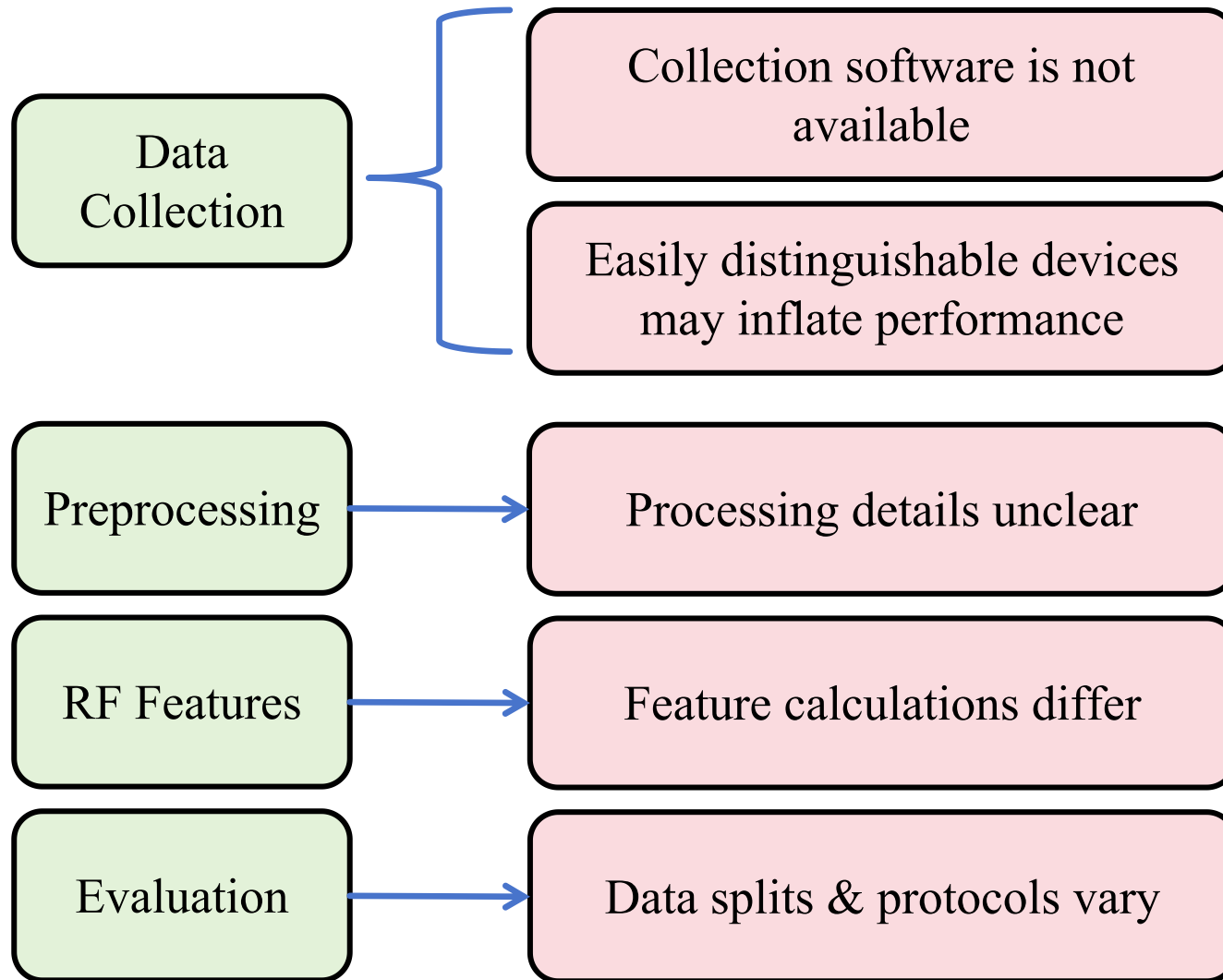
A simplified end-to-end RFF pipeline



Building an RFF experiment requires expertise across wireless systems, signal processing, and machine learning.



Reproducibility Challenges in RFF Research



We need a
reproducible RF
fingerprinting
system!

Learning Objectives

1. Understand the Wi-Fi RFF data acquisition process
2. Build an end-to-end RFF-based identification pipeline
3. Extract and visualize RF features
4. Train and evaluate a baseline model
5. Debug common implementation issues
6. Apply the workflow to your own datasets